Purdue Undergraduate Degree Planner

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Problem Statement

We will create a schedule planner for Purdue undergraduate students in this project. Even though myPurduePlan provides a platform for students to see courses that they have taken and the requirements for graduation, there is no easy way to visualize a comprehensive plan of study that spans the duration of their college career from that platform. Currently, students must either create their own plan by manually researching course details, such as requirements satisfied, credit hours, and prerequisites, or use the UniTime Scheduling Assistant, which is not designed for planning future semesters. Our degree planner stands out by allowing undergraduate students to effortlessly view and edit their comprehensive degree plan, providing instant access to information such as prerequisites and credit hours, all automatically populated and verified. Students would be able to create groups to share their schedules and check for common classes. Further, schedule owners can edit access permissions to authorize other users to copy the schedules as templates. By addressing the limitations of myPurduePlan and introducing user-friendly features, our degree planner simplifies the course-planning process by maximizing efficiency, making it a stellar solution for Purdue undergraduate students to plan their perfect course of study.

Project Objectives

1. Create an easy-to-use and visually pleasing user interface for planning and displaying one’s schedule throughout their academic journey in college. All of the software’s

functionality must be usable by all students, regardless of their studied degree or degree progress.

2. Help students monitor the progress of their requirements. By presenting students with facts about their completed coursework as well as unfulfilled degree requirements, they can make informed decisions when selecting courses.
3. Develop a group system for sharing schedules and finding common courses.
4. Create an authorization system for schedules that allows others to use the schedule as templates.
5. Our project will not aim to curate “suggestions” for the user’s degree, as any analyses we may conduct to do so will invariably be biased in their design. The application will gather and present all the necessary information to assist the user in making a decision instead of making decisions for them.
6. The application must not only protect the privacy of its users by requesting express permission from the user to store and retain the information about their degree progress, but also by ensuring that students’ data is only accessible and shareable by the student who submitted it.

Stakeholders

1. Users: Purdue Undergraduate Students, and Purdue Academic Advisors
2. Developers: Hamilton Wang, Jenny Chen, Joanna Lin, Mark Tseng, William Hou, Zoey Lu
3. Project Managers: Aline Itzel Becerra Carranza
4. Project Owners: Hamilton Wang, Jenny Chen, Joanna Lin, Mark Tseng, William Hou, Zoey Lu

Deliverables

1. Framework:

The project will have its web frontend made in React JS and its backend in Python and/or Java and Firebase. We will utilize Purdue.io to serve as the course information database.

2. Output:
    a. A comprehensive and systematic user interface utilizing React to render information easily accessible.
    b. A robust Firebase-based backend system to achieve a centralized course data center and user account information.
    c. This tool enables students to efficiently organize and create a personalized degree plan and allows academic advisors to better assist students during the course planning process. We hope our tool can be integrated with the existing course request system and make degree planning more convenient.